

Shuairu Zhu

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PERSONAL INFORMATION

Ph.D. in Astrophysics
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Research Interests: High-Redshift Galaxies, Lyman-Continuum Escape, Cosmic Reionization, Galaxy Evolution, HST/JWST Imaging and Spectroscopy

EDUCATION

Shanghai Astronomical Observatory, Chinese Academy of Sciences	Shanghai, China
Ph.D. in Astrophysics	2020–Present
Supervisor: Dr. Zhenya Zheng	
Thesis: Studies of High-Redshift Galaxies and Related Data Analysis Based on Space Telescope Observations	
Central China Normal University	Wuhan, China
B.S. Physics	2015–2019

PUBLICATIONS

15 papers/manuscripts in total, including 5 first-author papers/manuscripts. All papers are available in this [NASA/ADS library](#).

First-author papers and manuscripts

1. **Zhu, S.**, Zheng, Z.-Y., Bian, F., et al. 2026, *LCEz4-M1: A Lyman Continuum Emitter Candidate at $z = 4.444$ in the MUSE Hubble Ultra Deep Field*, *ApJL*, 1005, L3.
2. **Zhu, S.**, Zheng, Z.-Y., Yuan, F.-T., et al. 2025, *Lyman Continuum Leakers at $z > 3$ in the GOODS-S Field: Mergers Dominated*, *ApJL*, 982, L58.
3. **Zhu, S.**, Yuan, F.-T., Jiang, C., et al. 2024, *Lyman Continuum Leakers at $z > 3$ in the GOODS-S Field: Starburst or Not?*, *ApJL*, 974, L20.
4. **Zhu, S.**, Zheng, Z.-Y., Rhoads, J., et al. 2024, *The Hubble Deep Hydrogen Alpha (H δ H α) Project. I. Catalog of Emission-line Galaxies*, *ApJS*, 271, 5.
5. **Zhu, S.**, Zheng, Z.-Y., et al., *Analysis of HST ACS and WFC3 Calibration Fields for Future Space Astronomy Missions*, under review at *Astronomical Techniques and Instruments*.

Contributing-author papers and manuscripts

1. Zheng, Z.-Y., Xu, C., Liu, X., Chen, Y.-H., et al. 2026, *MCI: Multi-channel Imager on the Chinese Space Station Survey Telescope*, *Research in Astronomy and Astrophysics*, 26, 055020.
2. Yuan, F.-T., Zheng, Z.-Y., Jiang, C., & **Zhu, S.** 2026, *Two Lyman Continuum Escape Mechanisms at Play in $z \sim 0.3$ Galaxies Revealed by Infrared Observations*, under review at *ApJ*, [arXiv:2603.01472](https://arxiv.org/abs/2603.01472).
3. Lin, R., Zheng, Z.-Y., Jiang, C., Yuan, F.-T., et al. 2025, *Discovery of Local Analogs to JWST's Little Red Dots*, *ApJL*, 980, L34.
4. Chen, K., **Zhu, S.**, Jiang, L., & Zheng, Z.-Y. 2025, *Detection of Emission Line Galaxies in the Slitless Spectra of HST and CSST*, *Research in Astronomy and Astrophysics*, 25, 025015.
5. Yuan, F.-T., Zheng, Z.-Y., Jiang, C., **Zhu, S.**, et al. 2024, *Merging Signatures in an Offset Lyman Continuum Emitter at Redshift 3.8*, *ApJ*, 975, 53.
6. Lin, R., Zheng, Z.-Y., Yuan, F.-T., Wang, J.-X., Jiang, C., et al. 2024, *Intermediate-mass Black Holes in Green Pea Galaxies (IMBH-GP) I: A Candidate Sample from LAMOST and SDSS*, *Science China*

Physics, Mechanics & Astronomy, 67, 109811.

7. Ji, X., Zheng, Z.-Y., Wu, Q., Lin, R., et al. 2023, *Strange Quasar Candidates with Abnormal Astrometric Characteristics from Gaia EDR3 and SDSS (SQUAB-II): Optical Identifications*, *MNRAS*, 524, 1909–1926.
8. Lin, R., Zheng, Z.-Y., Wang, J.-X., Yuan, F.-T., et al. 2023, *Discovery of Five Green Pea Galaxies with Double-peaked Narrow [O III] Lines*, *MNRAS*, 524, 2224–2236.
9. Lin, R., Zheng, Z.-Y., Hu, W., Jiang, C., et al. 2022, *On the Origin of the Strong Optical Variability of Emission-line Galaxies*, *ApJ*, 940, 35.
10. Yuan, F.-T., Zheng, Z.-Y., Lin, R., **Zhu, S.**, et al. 2021, *CDFS-6664: A Candidate of Lyman-continuum Emission at $z \sim 3.8$ Detected by the Hubble Deep UV Legacy Survey*, *ApJL*, 923, L28.

RESEARCH SKILLS

Observing Experience: CTIO visiting astronomer, Blanco 4 m Telescope, Cerro Tololo, Chile, three half-night service observations; Lijiang 2.4 m Telescope, Gaomeigu, China, one full night.

Proposal Writing: Magellan Telescopes (M2FS, IMACS); CFHT (SITELE); Palomar Hale 5 m (DBSP); HST (Archival Program, WFC3).

Data Reduction & Analysis: HST (ACS, WFC3); CSST (MCI prototype); JWST (NIRSpec, NIRCам); VLT (MUSE, VIMOS); Magellan (IMACS).

Software: DrizzlePac, SWarp, SExtractor, PSFEx, GALFIT, CIGALE, BAGPIPES, TOPCAT, SAOImage DS9, pyspeckit, msaexp, grizli.

Languages: English (fluent); Chinese Mandarin (native).

SUPERVISING & PROFESSIONAL SERVICE

Assisted supervisor, Summer Research Program Shanghai, China, Jun.–Aug. 2023

- Student: Jing Wu (supervisor: Zhenya Zheng)
- Project: Upper Limits on the Contribution of Quasars to Cosmic Reionization

Member of LOC:

- 2025 Academic Symposium on Calibration Systems for Space- and Ground-Based Optical/Infrared Astronomical Telescopes, a two-day conference with 100+ participating experts and scholars

Referee for *The Astrophysical Journal*

APPROVED OBSERVING PROPOSALS

Magellan 6.5 m:

1. **Co-I** (CNTAC PI: Felipe Barrientos) 2024B
Spectroscopic Confirmation of HST Selected Lyman-Continuum (LyC) Leaking Galaxy Candidates at $z \sim 3-4$ in CDFS

CFHT 3.6 m:

1. **Co-I** (TAP PI: Z.-Y. Zheng) 2024A
Tracing the CGM and the LSS at $z = 3.2$ around the double-core Lyman- α quasar

Palomar Hale 5 m:

1. **Co-I** (TAP PI: R.-Q. Lin) 2021B
On the origin of the emission-line variability of emission-line galaxies

AWARDS & SCHOLARSHIPS

Total monetary awards: 379,000 CNY (approximately 54,500 USD).

National Graduate Student Award (30,000 CNY), Ministry of Education of PRC	2025
Keyuan Outstanding Student Award (5,000 CNY), CAS Shanghai Branch	2025
Best Oral Presentation Award (1,000 CNY), CSST Galaxy Evolution and Cosmology Symposium	2024
First Class Scholarship for Ph.D. Students (65,587.5 CNY per annum for 4 years), SHAO	2022–2026
Second Class Scholarship for Master's Students (40,325 CNY per annum for 2 years), SHAO	2020–2022
Merit Student, University of Chinese Academy of Sciences	2022, 2024, 2025

SELECTED PRESENTATIONS

Invited talks:

1. [The 2nd Graduate Academic Forum of SHAO, China](#) Mar. 2025
Lyman Continuum Leakers and Cosmic Reionization

Contributed talks:

1. [Scientific Workshop on In-Orbit Calibration of CSST MCI & IFS, China](#) May 2025
Ground Tests of the Survey Optical Facility–Multi-Channel Imager: Image Quality Results
2. [Escape of Lyman Radiation from Galactic Labyrinths, Greece \(accepted; presented by proxy\)](#) Apr. 2025
The high- z LyC leakers in the GOODS-S
3. [CSST Galaxy Evolution and Cosmology Symposium \(best oral presentation award\), China](#) Nov. 2024
The high- z LyC leakers in the GOODS-S
4. [2024 SHAO–XMU Astrophysics Academic Symposium, China](#) Nov. 2024
Lyman Continuum Leakers at $z > 3$ in the GOODS-S Field: Starburst Activity or Merger-Dominated Systems?
5. [2024 Symposium on Critical Questions in CSST Galaxy Studies, China](#) May 2024
The HST narrowband deep field survey and related sciences
6. [2024 Symposium on Frontiers in Galaxy Cosmology, China](#) May 2024
The Spectral Energy Distributions and Morphology of Lyman Continuum Leakers at High- z in the GOODS-S
7. [Space-Telescope-Era Galaxy Workshop, China](#) Nov. 2023
The Hubble Deep Hydrogen Alpha Project and Two Lyman Continuum Leaker Candidates at $z \sim 4.4$
8. [CSST Flux Calibration Workshop, China](#) Nov. 2023
Statistical Analysis of the Hubble Space Telescope Calibration Project

Contributed posters:

1. [The Inaugural Cosmic Frontier Center Conference \(poster flash talk\), US](#) May 2025
The high- z LyC leakers in the GOODS-S
2. [Kochel Cosmic Lyman Alpha Workshop \(poster flash talk\), Germany](#) Oct. 2024
The Hubble Deep Hydrogen Alpha Project and Two Lyman Continuum Leaker Candidates at $z \approx 4.4$

Summer schools:

1. [CSST Summer School, China](#) Jul. 2024
Galaxy science with the China Space Station Telescope
2. [18th Heidelberg Astronomy Summer School, Germany](#) Sep. 2023
Unraveling Galaxy Evolution with JWST

REFERENCES

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Dr. James Rhoads

NASA Goddard Space Flight Center

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Prof. Felipe Barrientos

Instituto de Astrofísica, Universidad Católica de Chile

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